

NANYANG JUNIOR COLLEGE
JC 2 PRELIMINARY EXAMINATIONS
Higher 2

CANDIDATE
NAME

MARK SCHEME

CLASS

BIOLOGY

9744/02

Paper 2 Structured Questions

13 September 2018

Candidates answer on the Question Paper.

No Additional Materials are required.

2 hours

READ THESE INSTRUCTIONS FIRST

Write your name and CT on all the work you hand in.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, highlighters, glue or correction fluid.

DO **NOT** WRITE IN ANY BARCODES.

Answer **all** questions in the spaces provided on the Question Paper

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's Use

1

2

3

4

5

6

7

8

9

10

Total

This document consists of **24** printed pages.

[Turn over

Answer **all** the questions in this section.

- 1 (a) A clinic sequenced the β -globin gene locus of five different patients and tabulated the results in the following table.

Table 1. Genetic profile of the 6th codon of patients' β -globin gene

Patient	DNA codon sequence*	Change in amino acid
1	GAG	Glu (unchanged)
2	GAA	Glu (unchanged)
3	GTG	Glu $\rightarrow$ Val
4	GAC	Glu $\rightarrow$ Asp
5	GTG	Glu $\rightarrow$ Val

* DNA sequence on the 6th codon of the human β -globin gene

Based on the information in **Table 1**,

- (i) Which patient(s) has/have the sickle-cell anaemia mutation?

Patients 3 & 5.

[1]

- (ii) Explain how the change of amino acid would result in a sickle shaped red blood cell.

Glutamic acid and valine are amino acids with very different properties. Glutamate is hydrophilic while valine is hydrophobic;

At low oxygen concentration, hydrophobic areas of different molecules would stick together, resulting in HbS molecules polymerizing/ precipitating into fibres;

The long fibres distort the membrane of the red blood cell giving it its distinct sickle shape.

[3]

- (iii) Suggest and explain why patient 4 does not have sickle cell anaemia.

Both glutamic acid and aspartic acid have the same chemical property [they are both are polar and charged amino acids];

Hence, the change to the structure of the β -globin protein and eventually haemoglobin is insignificant.

[2]

Fig. 1 shows a linear chromosome undergoing the first round of DNA replication. The arrows show the direction of synthesis for the daughter DNA strands. Ends A, B C and D represent the ends of the newly synthesized daughter DNA strands.

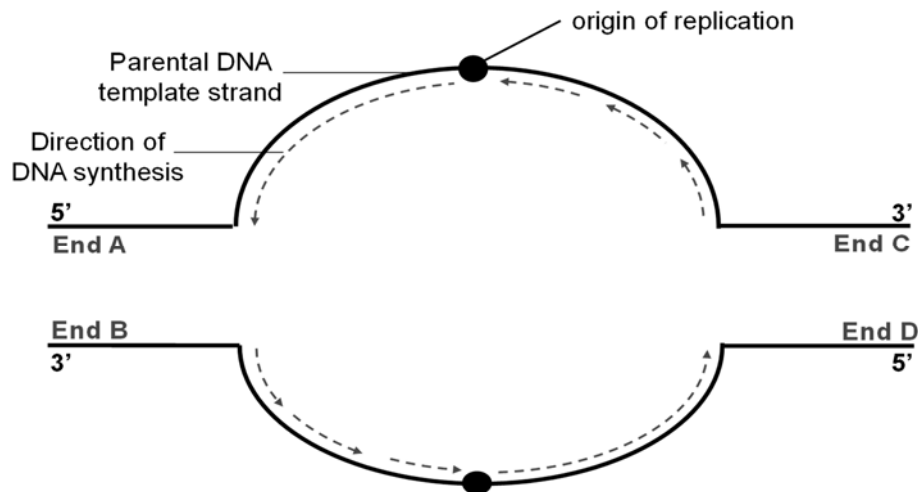


Fig. 1

(b) With reference to **Fig. 1**,

- (i)** state which end(s) will experience the end-replication problem after the first round of DNA replication has been completed.

End C and End B

[1]

- (ii)** suggest one modification that a researcher could do to the parental DNA strands in order to completely remove the end-replication problem.

Ligate/join the 5' and 3' ends of each parental strand together;
OR Circularise the linear DNA chromosome.

[1]

- (c)** Telomerase is an enzyme that is found in certain cell types and it helps to ameliorate the end replication problem by lengthening the ends of the chromosomes, which are also known as telomeres.

Explain why the telomerase enzyme does not prevent the end-replication problem from occurring despite being able to lengthen the telomeres.

When the RNA primer is being removed at the 5' ends of the newly synthesized DNA.

There is still no 3'OH end at the 5' end for DNA Polymerase to add free deoxyribonucleoside triphosphates to.

[2]

[Total: 10]

- 2 Fig. 2.1 shows some *Allium* sp. plant cells in various stages of the mitotic cell cycle.

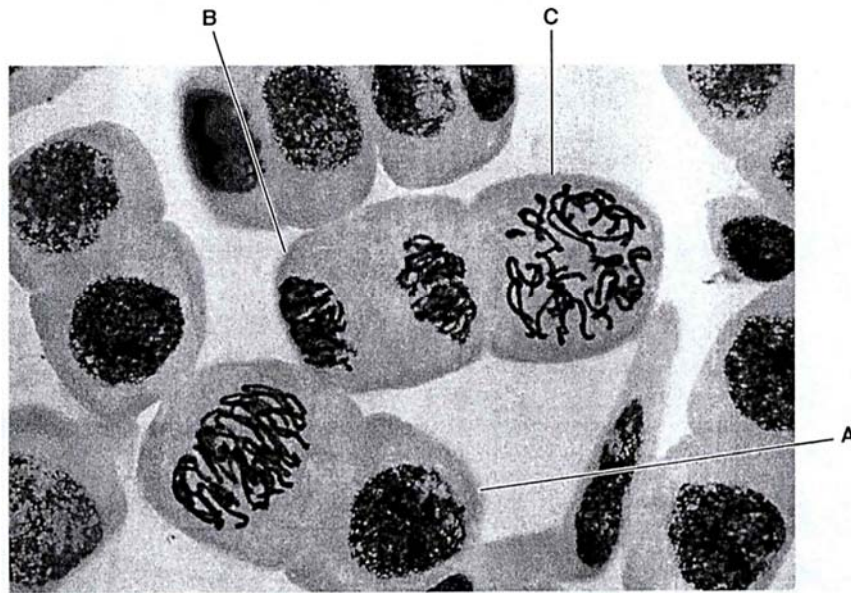


Fig. 2.1

- (a) (i) Identify the three stages shown by the labelled cells.

A Interphase

B Anaphase

C Prophase

[3]

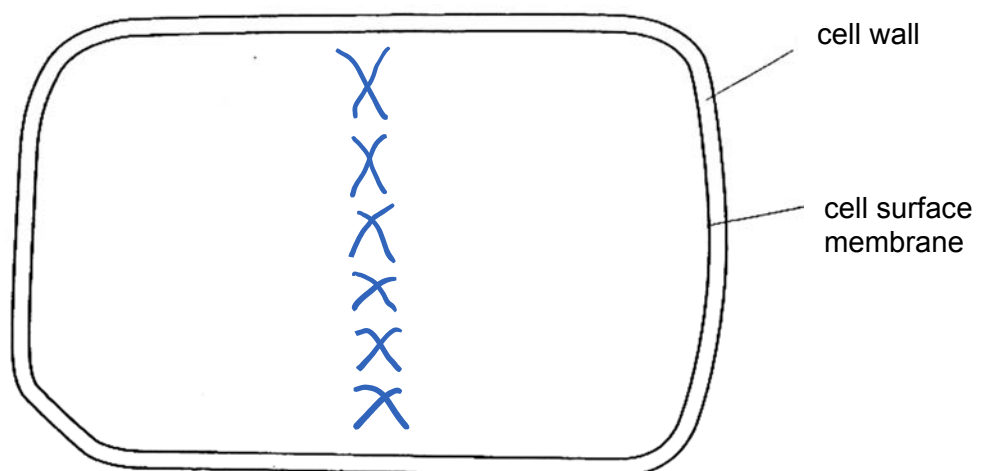
- (ii) Identify the stage of mitosis that follows that shown in cell C.

Metaphase

[credit will be given as long as the stated answer follows their answer in a(i)C]

[1]

- (iii) In the cell outline below, draw and label the structures visible in a cell that is in the stage you have named in (ii). $2n$ for this plant is 6.



Drawing within cell outline; label chromosomes; $2n=6$;

® asters / centrioles which are absent in plants [3]

- (b) Uncontrolled cell division can result in cancer. Some types of cancer can be treated by chemotherapy, which involves the injection of chemicals into the bloodstream.

One chemical used for chemotherapy is called Methotrexate. This is a reversible competitive inhibitors of one of the enzymes in the metabolic pathway that results in the formation of purines.

Explain how the use of Methotrexate will slow down the mitotic cell cycle.

(Due to competition, less purines formed) so less nucleotides synthesised ;

Leading to less DNA replication ; (slowing down mitotic cell cycle)

[2]

- (c) Prokaryotic organisms such as *Escherichia coli* divide by simple cell splitting (binary fission), not mitosis.

Apart from ribosomes, prokaryotes have no organelles comparable to those found in eukaryotes and have a circular 'chromosome' with no centromere.

With reference to the information above and your knowledge of mitosis, suggest why mitosis does **not** occur in prokaryotes.

Lack of centrioles / microtubules to separate the chromosomes during anaphase ;

Circular chromosomes does not allow for separation unlike linear chromosomes ;

[2]

[Total: 11]

- 3 The coat colour of Norwegian cattle is mainly determined by the distribution of two pigments: red and black. Both pigments are produced by the action of the enzyme tyrosinase in cells called melanocytes. Low enzyme activity leads to the production of red pigment, while high enzyme activity brings about black pigment production.

The activity of the enzyme is increased when melanocyte stimulating hormone (MSH) combines with a MSH receptor. The receptor is coded for by the gene, **R**, which has three alleles, **R^D**, **R^A** and **r**. **R^D** and **R^A** each codes for a receptor with a different activity. No receptor is produced by the recessive allele, **r**.

The dominant allele of a second gene, **B**, codes for a protein which binds to and blocks the MSH receptors coded for by **R^A**, thus preventing stimulation of tyrosinase activity in a melanocyte. The receptors coded for by **R^D** is insensitive to the protein coded by **B**. The recessive allele, **b**, does not produce a functional protein.

- (a) (i) State the name given to the interaction between the **R** and **B** gene loci.

Epistasis;

[1]

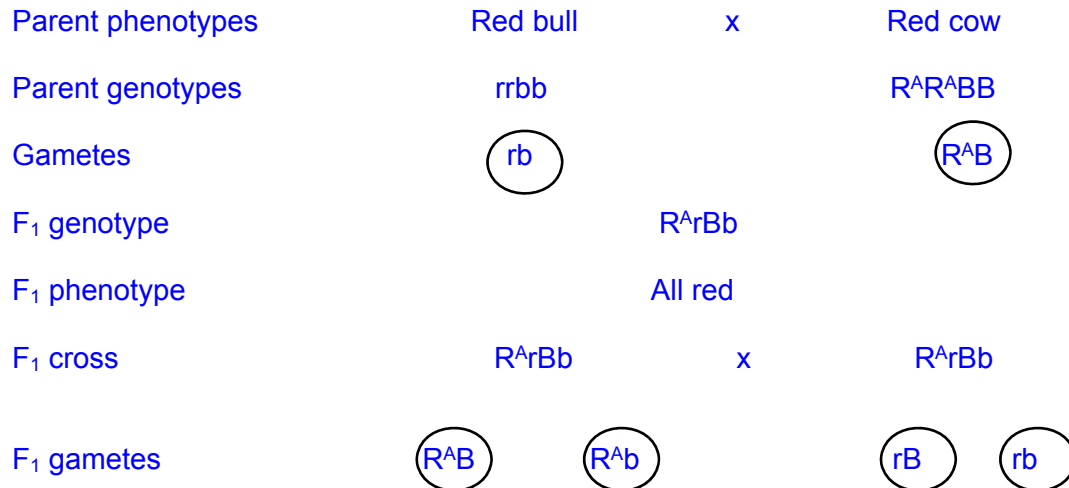
- (ii) Explain why animals with the genotype **R^AR^ABB** have red coats.

Inhibitor / Protein that blocks MSH receptor coded for/produced and hence MSH unable to bind to its receptor to increase/stimulate tyrosinase / enzyme activity;

[1]

- (iii) A red cow, with genotype $R^A R^A BB$ is mated with a bull which is homozygous recessive at both gene loci.

Draw a genetic diagram in the space below to show the expected genotypes and phenotypes and their ratios in the F_1 and F_2 generations.



Punnett Square:

	$R^A B$	$R^A b$	$r B$	$r b$
$R^A B$	$R^A R^A B B$ Red	$R^A R^A B b$ Red	$R^A r B b$ Red	$R^A r B b$ Red
$R^A b$	$R^A R^A B b$ Red	$R^A R^A b b$ Black	$R^A r B b$ Red	$R^A r b b$ Black
$r B$	$R^A r B B$ Red	$R^A r B b$ Red	$r r B B$ Red	$r r B b$ Red
$r b$	$R^A r B b$ Red	$R^A r b b$ Black	$r r B b$ Red	$r r b b$ Red

F₂ phenotypic ratio 13 red coat : 3 black coat

Mark allocation:

1. Correct parent gametes;
2. Correct F₁ phenotype and genotype;
3. Correct F₁ gametes;
4. Correct F₂ phenotypes and genotypes in Punnett square;
5. Correct F₂ phenotypic ratio;

[5]

During a health screening exercise of cattle in a farm, the height of the bulls was measured and the data collected is shown in **Table 3**.

Table 3

Height/cm	Number of bulls
131—135	3
136—140	9
141—145	21
146—150	12
151—155	2

(b) Distinguish between the two types of variation shown in coat colour and height in the Norwegian cattle.

1. * Coat colour shows discontinuous variation	While height shows continuous variation;
2. Discrete phenotypic classes and no intermediates are observed	A range of phenotypes are observed;
3. Coat colour is controlled by one or two major genes, which may have two or more allelic forms.	Height is controlled by a large number of genes (polygenes);
4. Effect of individual genes can be observed.	Effect of individual genes cannot be observed;
5. Effect of genes is not additive.	Effect of genes is additive;
6. The environment has a small effect on the phenotype.	Environment has a large effect on the phenotype;

* Compulsory point;

[3]

In the honey bee colony, the queen bee is solely responsible for laying eggs and the drones for fertilizing her. The worker bees have well-developed mouthparts and structural adaptations for collecting nectar and pollen to gather food and to perform other duties in the hive. Male bees are developed from haploid eggs while both queen and worker bees develop from fertilized eggs.

(c) Explain how the phenotypic differences between the queen and the worker bees come about despite both being developed from fertilized eggs.

1 The different diets brought about differences in gene expression in cells, resulting in the phenotypic differences;

2 Larvae fed on diet of royal jelly throughout development become queen bees while those fed on worker jelly become worker bees;

[2]

[Total: 12]

4 Fig. 4.1 shows a cyclic series of reactions that occurs during photosynthesis.

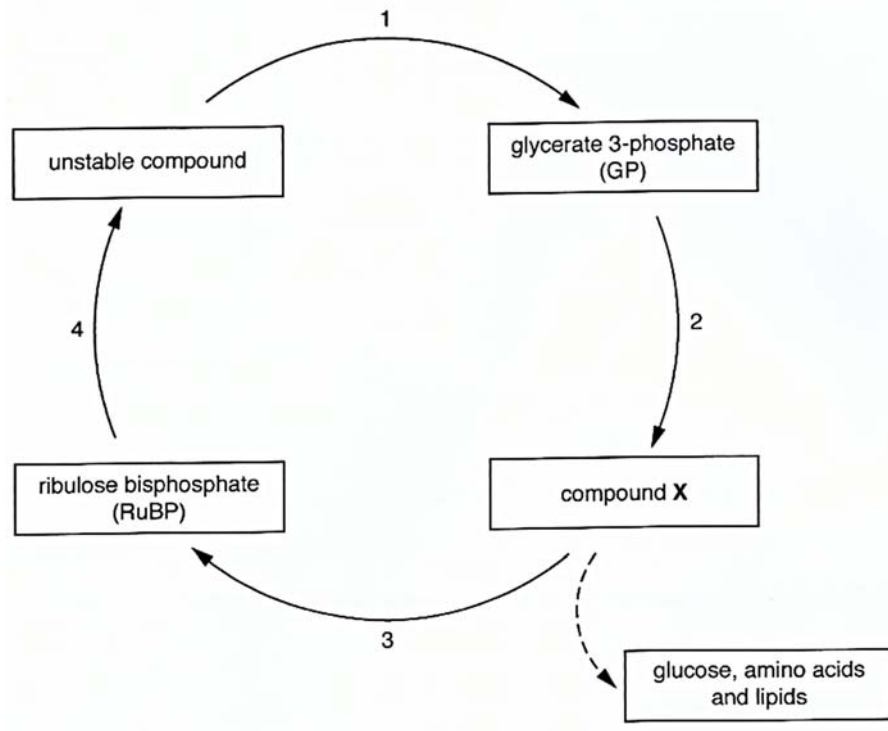


Fig. 4.1

(a) (i) State the name of this cycle.

Calvin cycle / dark rxns / light independent rxns

[1]

(ii) State the number of carbon atoms in one molecule of compound X.

3

[1]

(iii) Give the label number of each solid arrow that represents reactions where energy is released from ATP.

2 and 3

[1]

(iv) Give the label number of the solid arrow that represents reactions where carbon dioxide is taken up.

4

[1]

(b) Fig. 4.2 is an electromicrograph of a chloroplast.

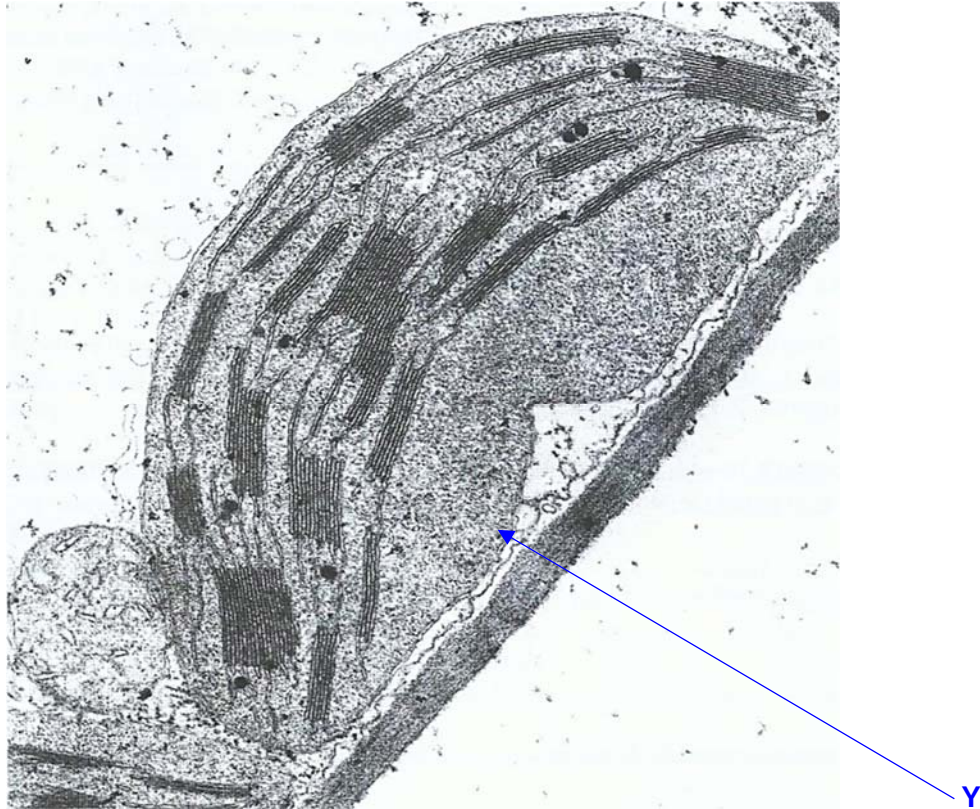


Fig. 4.2

- (i) On Fig. 4.2 use the letter Y and a label line to identify where the cyclic series of reactions shown in Fig. 4.2 occurs. (stroma)

[1]

- (ii) Chloroplasts and mitochondria both have double membranes.
State two other structural similarities between chloroplasts and mitochondria.

Have ETC and stalked particles on their membranes ;

70S ribosomes

Circular DNA

[2]

(c) The enzyme rubisco has an important role in the cycle of reactions shown in **Fig. 4.1**.

Both oxygen and carbon dioxide can bind to the active site of rubisco.

- (i) With reference to the information above, suggest why an increase in carbon dioxide concentration increases the rate of reactions shown in **Fig. 4.1**.

Competitive inhibition

.....
Increase in CO₂ will cause rubisco to bind more frequently with CO₂ than O₂

.....
Increasing the chances of formation of CO₂-rubisco complex

.....
Increase in rubisco increases rate of reactions in fig.4.1 as it is a cyclic process ;

[2]

- (ii) At low temperatures, the activity of rubisco is low and this limits the rate of photosynthesis. Some plants that have a high rate of photosynthesis at low temperatures have an enzyme, **Z**, which increases the activity of rubisco.

Suggest how scientists could make use of this information to improve a crop plant so that it could be grown in cold conditions.

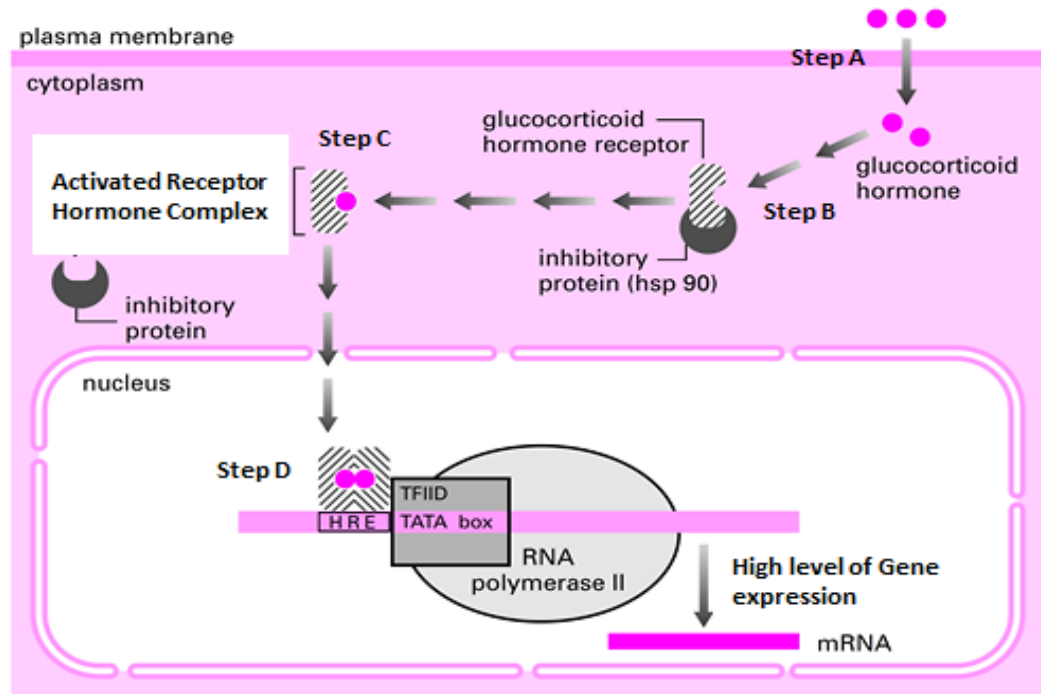
Genetic engineering

.....
Transfer the gene coding for enzyme Z into crop plant

.....
[2]

[Total: 11]

- 5 Glucocorticoids are steroid hormones produced by the adrenal cortex that increase the transcription of several genes important in carbohydrate and protein metabolism. **Fig. 5.1** below shows how glucocorticoids can pass through the plasma membrane to enter the cytosol and bind to glucocorticoid hormone receptors. In the absence of glucocorticoid hormone, its receptor remains in the cytosol and is inactive.



Hormone response element (HRE)

Fig. 5.1

- (a) (i) Glucocorticoid hormone receptor is a class of transcription factors. With reference to **Fig. 5.1**, explain how receptor hormone complex (Step C) can be activated.

Being a steroid hormone, glucocorticoids are hydrophobic and can pass through the hydrophobic core of the phospholipid bilayers ;

It binds to hormone receptor and remove inhibitory protein (hsp90) thereby activating the receptor hormone complex ;

@ Ref to binding of glucocorticoid hormone to inactivated glucocorticoid receptor

@ which leads to removal of Hsp90 protein forming the glucocorticoid receptor-hormone complex

[2]

- (ii) Suggest how glucocorticoid hormone receptor is able to bind to the specific region of the DNA known as HRE site shown in step D of **Fig. 5.1**.

Activated receptor hormone complex enter nucleus via nuclear pores ;

Dimerise ;

The dimer has a dna binding domain that can recognise the base sequence of HRE site ;

Complementary in terms of shape size charge and orientation ;

- This occurs via Specific DNA - protein interactions which depend upon the sequence of bases in the DNA. [1]
- These DNA - protein interactions are mediated by the following bonds [1]:
 - Hydrogen bonding
 - Ionic interactions: Salt bridges; R groups of protein - DNA backbone interactions

Other forces: van der Waals, hydrophobic

[2]

- (iii) With reference to **Fig. 5.1**, explain how high level of gene expression may be regulated by activated receptor hormone complex within the nucleus.

With the binding of dimerised activated receptor hormone complex to HRE site, it recruits TFIID to bind to tata box of promoter ;

Which recruits RNA polymerase II ;

Stabilises the transcription initiation complex ;

Increasing rate of transcription ;

Forming more mRNA → translation of proteins → high levels of gene expression

- The glucocorticoid receptor–hormone complex acts as an activator which binds to the HRE which is an enhancer sequence / distal control element.
- The activators bind to certain general transcription factors such as TFIID and mediator proteins / Recruits basal Transcription factors
- Stabilises / increases the binding affinity of RNA polymerase II to TATA Box / Promoter sequence, and increase the rate of formation of stable / stability of transcription initiation complex for RNA synthesis to begin.

[3]

Fig. 5.2 below shows the various steps involved in the processing of primary RNA transcript.

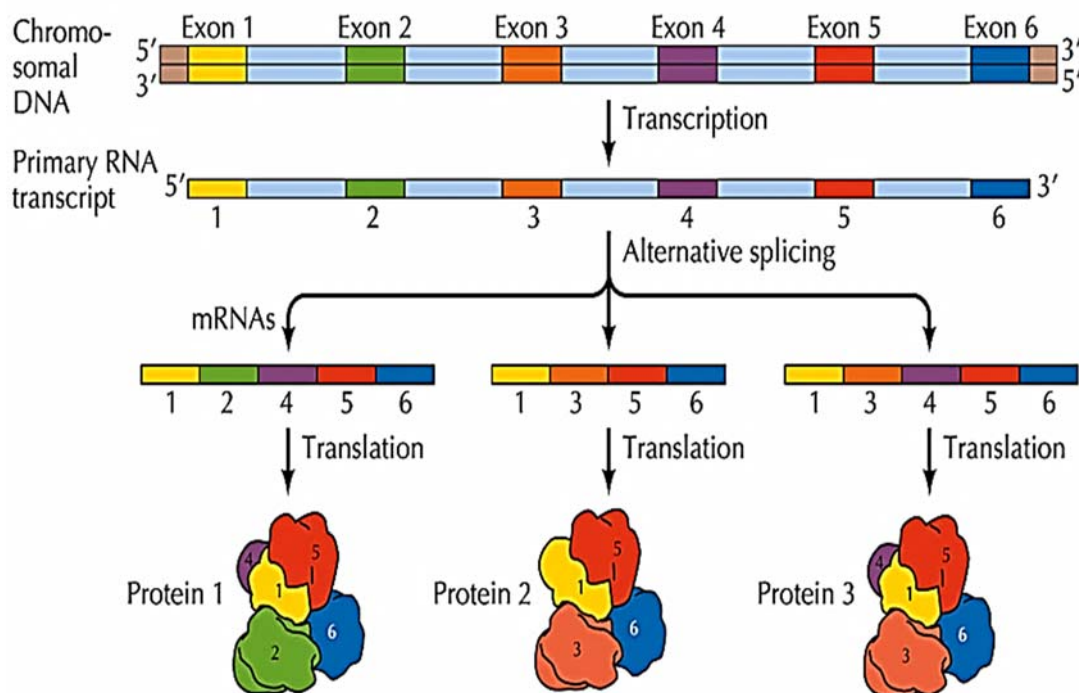


Fig. 5.2

- (b) (i) Explain why alternative splicing is essential in eukaryotic cells from an evolutionary view point.

Allow more proteins to be translated from a smaller number of genes

Proteins can result in change in phenotypes which provide more variation to cope with the ever changing environments

Allow for smaller genomes to be stored in cells

1. This process allows a single gene to code for different types of proteins / This allows eukaryotic genome to direct the synthesis of many more proteins than would be expected from its fixed protein-coding genes
2. Particular exons of a gene may be included within, or excluded from, the final, processed messenger RNA (mRNA) produced from that gene.
3. Ref to proteins translated from alternatively spliced mRNAs will contain differences in their amino acid sequence and, often, in their biological functions leading to variations to occur in a population. e.g. phenotypes etc.

(Accept either point 2 or 3, but not both together.)

[3]

- (ii) Prior to completion of alternative splicing, the primary RNA is modified in several important ways. Explain how **one** of these ways helps to regulate gene expression.

5' capping – facilitate export to cytoplasm and ribosomal binding and stability

3' poly A tail – regulate half life of mature mRNA and hence amount of proteins formed

Any one of the following:

- Primary mRNA is capped by the addition of 7-methylguanosine cap to 5' end of the transcript to protect the growing RNA transcript from degradation by RNases.

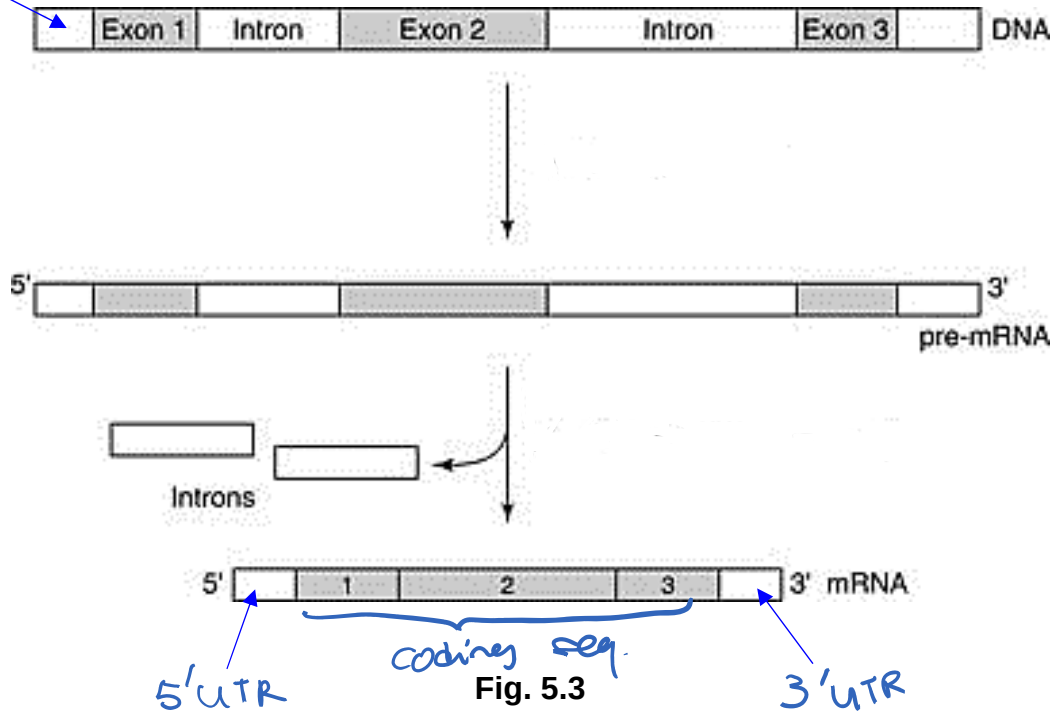
A poly-A tail is added to the 3' end by the enzyme poly-A polymerase. The poly-A tail has several functions: (1) It protects against RNases/ exonuclease and therefore increases the stability of mRNA molecules in the cytoplasm, (2) both it and the 5' cap are required for transit through the nuclear pore from the nucleus to cytoplasm, and (3) it increases the efficiency of translation on the ribosomes.

[1]

(iii) With reference to Fig. 5.3, label the following region (label only once):

- (i) 5' UTR
- (ii) 3' UTR
- (iii) coding sequence
- (iv) promoter

Promoter



[1]

[Total: 12]

6 Viruses employ various mechanisms to evade the host immune response.

The presence of Major Histocompatibility Complex (MHC) class I and class II molecules on Human Immunodeficiency Virus (HIV) has been shown to render the virions more infectious.

(a) Suggest how HIV acquires the MHC molecules.

1. The new HIV acquires its outer envelope when it buds off
2. from the plasma membrane of immune cell (macrophage, dendritic cells, CD4+ T cells) which has MHC Class I and II embedded.

[1]

Fig. 6.1 shows the number of T helper cells in the blood and the number of HIV viruses in the body over the course of an untreated HIV infection.

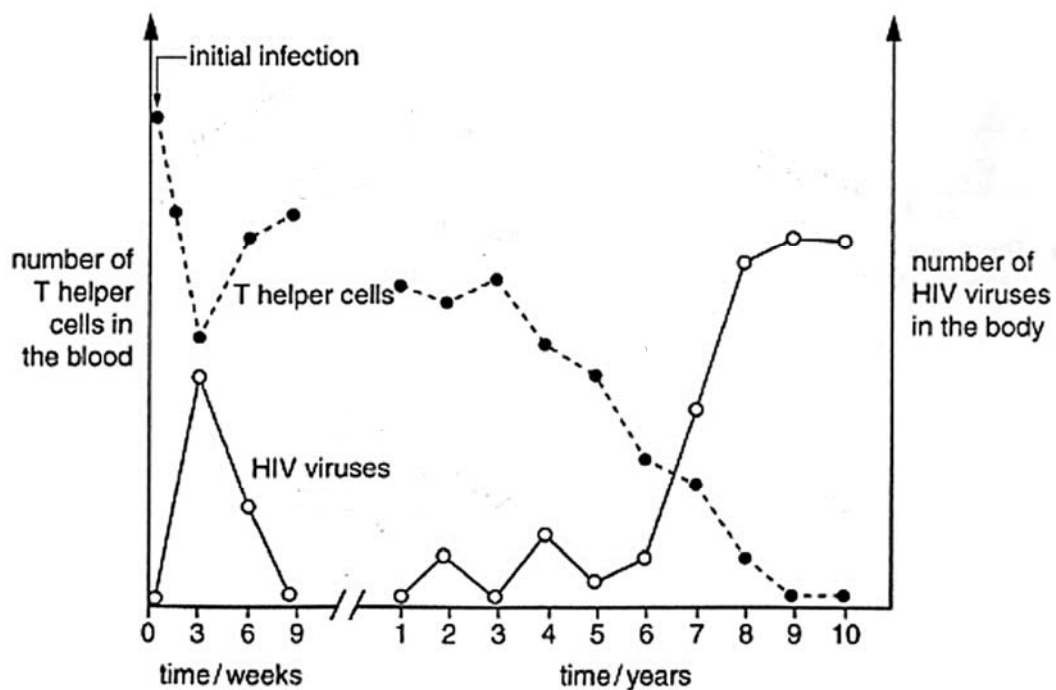


Fig. 6.1

(b) (i) Describe the changes shown in **Fig. 6.1**.

From 0 to 3 weeks, number of HIV viruses in the body increases and number of T helper cells in the blood decreases;

From 3 to 9 weeks, number of HIV viruses in the body decreases and number of T helper cells in the blood increases (still lower than original number at week 0);

From 1 year to 10 years, number of HIV viruses in the body increases and number of T helper cells in the blood decreases;

[3]

(ii) Suggest how the changes in the number of T helper cells shown in **Fig. 6.1** would affect the health of an untreated HIV-infected individual over the course of the infection.

Overall trend over 10 years:

there is a general decrease in number of T helper cells as the infected T helper cells are killed by the cytotoxic T cells / infected T-cells dies as more HIV virions bud off;

0-9 weeks: Flu-like symptoms due to innate immune responses mounted upon infection;

From 9 weeks – 3 years, despite a fall in number of T helper cells, there are relatively high levels of T helper cells which still allows for the immune system to manage the free viruses and infected cells, therefore infected person is asymptomatic;

From 3 years to 10 years, the drastic decrease in T helper cells below critical levels result in the loss of adaptive immunity/ cell-mediated immunity and onset of Acquired Immune Deficiency Syndrome, causing individuals to be vulnerable to opportunistic infections;

[3]

(c) Although a person may be infected only once by one strain of HIV he may eventually have many strains of HIV. Briefly explain why.

- 1 Different strains of HIV are genetic variation that arise via mutations;
- 2 HIV mutates very rapidly;
- 3 Reverse transcriptase is inaccurate in its replication / Unlike DNA polymerase, reverse transcriptase does not proof read the nucleotides added;
- 4 Reproductive rate of rate of HIV is high $\propto$ increases chances of mutant strains arising;
- 5 Mutation rate is greater because the genetic material is single stranded (RNA);
- 6 RNA single stranded $\rightarrow$ less stable because no complementary strand $\rightarrow$ no template for DNA repair;

Examiners' comments: Most students failed to appreciate that different strains refer to genetic variants, instead misunderstanding that different strains referred to many copies that were genetically identical. Another common mistake was to misread the question – even though the question clearly stated a person is “infected only once”, some students explained the presence of several strains due to opportunistic infections (by other strains). Some students were aware that mistakes in replication led to mutations, but failed to understand that this was due to the inherent nature of reverse transcriptase. A few also mentioned that RNA was less stable, but did not realise that it was the nature of being single-stranded rather than being RNA per se, that led to less stability.

[3]

[Total: 10]

- 7 (a) In bacteria, it is common for two or more genes to be arranged together in an operon.
 (i) Briefly discuss the arrangement of genetic sequences within an operon.

Any 1 from below (max. 1 mark):

- Flanked by a promoter to signal the beginning of transcription and a terminator to signal end of transcription
- An operator site, is usually present, to serve as binding site for regulatory proteins called activators or repressors

1 from below:

Two or more structural genes that encode different proteins are found between these two sequences

[2]

- (ii) State the biological advantage of an operon organization.

Co-ordinately regulate a group of genes whose encoded proteins have a common function to reduce wastage of resources / response to environmental changes quickly

[1]

- (b) The sequential use of two sugars by a bacterium is known as diauxic growth. It is a common phenomenon among many bacterial species. When glucose is one of the two sugars available, it is typical that the bacterium metabolises glucose first, and then a second sugar after the glucose has been used up. Among *E. coli* and related species, diauxic growth is regulated by intracellular cAMP levels and the catabolite activator protein (CAP).

- (i) Summarise the effects of glucose and lactose on the ability of the lac repressor and the cAMP-CAP complex to regulate the lac operon.

- Lactose present, operon is switched on, as the repressor protein is removed from the operator site
- Glucose present, operon is turned off due to the dominating effect of the lac repressor protein

Lactose and glucose present, the expression of the *lac* operon is greatly decreased, the lac repressor is removed from operator site, and the catabolite activator protein is not bound to the CAP site, reducing RNA polymerase binding to begin transcription

[3]

You have isolated a mutant strain of *E. coli* in which the lac operon is constitutively expressed. To understand the nature of this defect, you create a merozygote (diploid cell) in which the mutant strain contains an F' factor with a normal lac operon and a normal lacI gene. You then compare the mutant strain and the merozygote with regard to their β -galactosidase activities in the presence and absence of lactose.

You obtain the following results:

	Addition of lactose	Amount of B-galactosidase (percentage of mutant strain in the presence of lactose)
Mutant	No	100
Mutant	Yes	100
Merozygote	No	100
Merozygote	Yes	200

Fig. 7

(ii) Explain the nature and effect of the defect in the mutant strain.

Mutation in the operator site

Repressor protein cannot bind to prevent lac operon expression in the absence of lactose

[2]

[Total: 8]

- 8 *Hebe* species are native shrubs found throughout New Zealand, particularly on the banks of lowland streams and rivers, as well as in the alpine (above the treelines) and subalpine zones of the mountains.

Table 8.1 shows the description of seven *Hebe* species.

Table 8.1

Species	Morphology	Habitat	Number of chromosomes in a gamete
<i>H. imbricata</i>	Rounded shrub up to 0.6 m tall.	Drier mountains of the South Island.	20
<i>H. raoulii</i>	Small low growing shrub up to 0.3 m tall.	Drier mountains and hills of the South Island.	21
<i>H. crenulata</i>	Low shrub up to 1 m tall.	Subalpine shrubland and grassland, often in shallow or rocky soils of northern South Island.	40
<i>H. elliptica</i>	Bushy shrub up to 2 m tall.	Seaside, west coast of South Island.	20
<i>H. cupressoides</i>	Shrub up to 2 m tall.	River-flats and terraces, as well as subalpine east of the South Island.	21
<i>H. gracillima</i>	Shrub up to 2 m tall.	Damp swampy places in the west coast of South Island.	40
<i>H. topiaria</i>	Neatly rounded bushy shrub up to 2 m tall.	Wet subalpine areas in the north of South Island.	61

It has been proposed that the ancient population that gave rise to the *Hebe* species in New Zealand most likely came from *H. cupressoides* in Australia about five million years ago.

The movement between the Australian Plate and Pacific Plate over the last five million years resulted in the formation of the Southern Alps. The rise of the Southern Alps caused the west coast of the South Island to be much wetter than the east coast. The map of New Zealand is shown in Fig. 8.1.

Between 1.6 million and 10,000 years ago, New Zealand was exposed to cold glacial and warmer interglacial periods. These caused the sea levels to fluctuate by up to 135 m. In the glacial periods, forests retreated from the mountains towards the coasts. This also changed the soil quality in the regions.

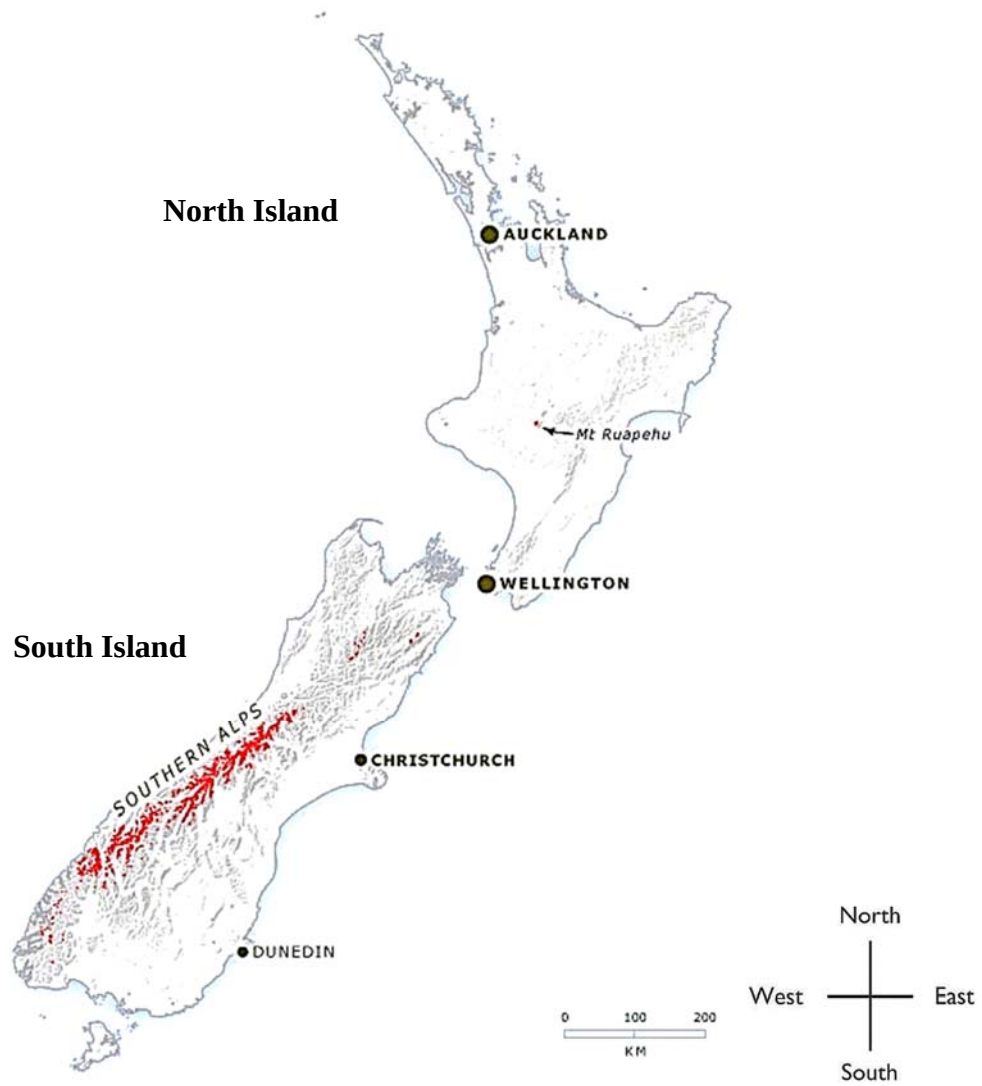


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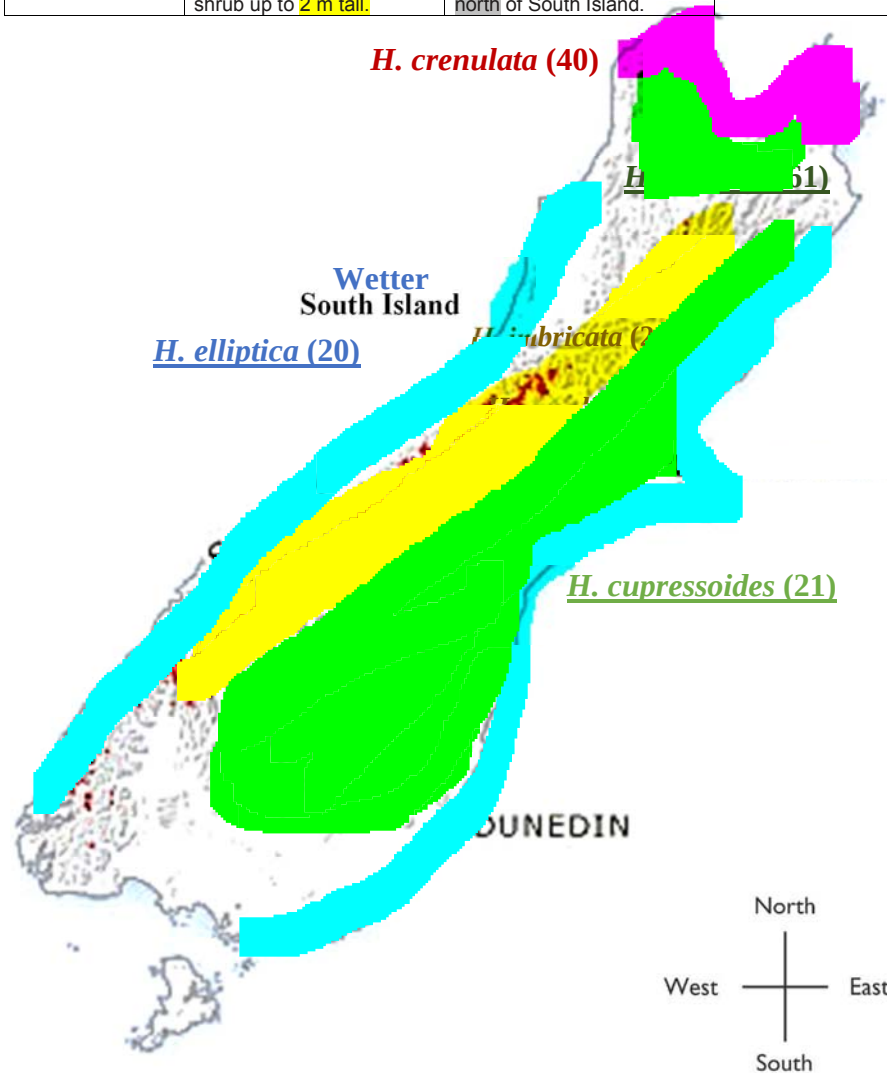


Fig. 2.1

(a) Explain why the population is the smallest unit that can evolve.

Evolution Lecture Notes p.2, 29

[Definition of population]

1. A population is a group of individuals of the SAME species
2. that live in the SAME area at the SAME time.

[Explain how natural selection operates]

3. Only the population can have genetic variation, i.e. different alleles contributing to the diversity of the gene pool that provides the raw materials for natural selection to act on the phenotype of individuals/ OWTTE.
4. In order for evolution to occur, the favourable alleles must be passed down to the next generation/ OWTTE.

[Otherwise the phenotype of the population will not change]

[Explain why population is the smallest unit to evolve]

5. The allele of an individual CANNOT change throughout its lifetime/ Individuals do NOT have genetic diversity for natural selection to select on/ OWTTE,
6. ONLY the allele frequencies of a population can change (microevolution).

[3]

(b) (i) Explain how the environment factors contributed to the formation of different species of *Hebe*.

1. [QF] The formation of Southern Alps, glacial and interglacial periods, fluctuation of sea levels resulted in different moisture content in the soil (west coast being wetter than the east)
2. [QF, Explain] which causes the change in soil quality.
3. [QF] The formation of Southern Alps also resulted in different climates/temperatures in different parts of the island (i.e. seaside, river flats, subalpine, mountains).
4. [QF, Explain] All of these act as selection pressures. For example, plants in the wetter region (e.g. *H. elliptica*, *H. cupressoides*, *H. gracillima*, *H. topiaria*) are likely to take up more water and nutrients from the soil, thus they are at a selective advantage and are able to grow taller/ better/ Accept Reverse Argument.
5. [QF, Explain formation of species] The formation of Southern Alps also forms a geographical barrier that resulted in geographical isolation of the different *Hebe* species.
6. [Explain formation of species] Thus, there is no gene flow between the species, they are unable to interbreed to produce viable and fertile offspring.
7. [Explain formation of species] As they live separately, they continue to accumulate numerous modifications/ adaptations, thus causing them to diverge to become new species over time.

[3]

(ii) Suggest how the 61 chromosomes of *H. topiaria* could have arisen.

[3]

- | | |
|---|---|
| <ol style="list-style-type: none"> 1. The spontaneous random mutations in the chromosomes of ancestral species of <i>H. cupressoides</i>/ The spindle fibres failed to attach to the kinetochore at the centromere of the chromosomes in the ancestral species of <i>H. cupressoides</i> | <ol style="list-style-type: none"> 1. The 61 chromosomes of <i>H. topiaria</i> could be hybrid due to the fertilisation between <i>H. cupressoides</i> (n=21) and <i>H. gracillima</i> (n=40) 2. The species are able to interbreed and produce viable offspring (n=61) |
|---|---|

2. resulted in polyploidy OR gain of extra set of chromosomes in the offspring
[Accept: triploidy]
3. [QF] one set of 21 chromosomes in gametes becomes three sets of chromosomes (63 chromosomes)
4. and then aneuploidy OR gain/ loss of 2 chromosomes.
5. [QF] 63 chromosomes in gametes becomes 61 chromosomes
Accept: Other chromosome numbers to arrive at 61 chromosomes in the gametes
6. This results in the formation of the species H. topiaria which is able to survive till reproductive age
7. and produces gametes with 61 chromosomes due to reduction division.
3. The spontaneous random mutations in the chromosomes of the hybrid/ The spindle fibres failed to attach to the kinetochore at the centromere of the chromosomes in the hybrid
4. resulted in polyploidy OR gain of extra set of chromosomes in the offspring
5. [QF] 61 chromosomes in the hybrid organism becomes 122 chromosomes in the hybrid organism.
6. This results in the formation of the species H. topiaria which is able to survive till reproductive age
7. and produces gametes with 61 chromosomes due to reduction division.

- (c) Suggest **two** reasons why **fossils** of **whole plant** are **difficult to obtain**, as compared to **animal fossils**.

Evolution Lecture Notes p.25

1. Plants lack hard tissues (e.g. bones) which are found in animal fossils/
OWTTE.
2. The soft tissues of plants also decompose rapidly even before they are fossilized as compared to animals.

[2]

[Total: 11]

- 9 The toxins released by some pathogenic bacteria can be altered chemically so that they are harmless. These harmless toxins are called toxoids.

Toxoids are used in vaccines to provide protection against some infectious diseases.

Describe the response of the immune system to the injection of a toxoid.

[5]

9700/21

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Question	Answer	Marks
5(a)	<i>five from:</i> A 'cells' for 'lymphocytes' throughout 1 ref. to antigen presentation ; A macrophage presents antigen 2 recognition / binding (in context of B-, or T-lymphocytes) ; e.g. cell surface receptor <u>specific</u> to toxoid / clonal selection A immunoglobulin / antibody on surface of B-lymphocytes 3 T-lymphocytes or B-lymphocytes, divide by mitosis / clonal expansion ; 4 (some) plasma cells, formed / AW ; R if in context of T-lymphocytes 5 antibody molecules / antibodies, secreted / produced / released ; R if in context of T-lymphocytes 6 T helper cells secrete, cytokines / interleukins ; 7 ref. to action of cytokines / interleukins ; e.g. to stimulate, humoral response / B-lymphocytes / to stimulate <u>macrophages</u> / angry <u>macrophages</u> 8 formation of memory cells ; <i>only in context of B- / T-lymphocyte activation</i>	5

[Total: 5]

- 10** Reef building corals are marine invertebrates found in shallow, clear tropical seas. The corals secrete an exoskeleton of calcium carbonate that becomes the underlying structure of the coral reef.

Zooxanthellae are a group of unicellular photosynthetic algae that live inside the cells of reef building corals. The relationship is beneficial to both the zooxanthellae and the coral.

- (a)** Evidence shows that the relationship between zooxanthellae and the reef building coral has evolved by free living algae invading corals that did not contain algae.
- (i)** Corals that do not need zooxanthellae can live at greater depth than reef building corals.

Explain why.

Any **three** from:

(corals without zooxanthellae) no reliance on light / ora ;

Reef building corals (have zooxanthellae so) need light (to photosynthesize) ;

As depth increases, less light penetration / more light absorbed (by water) ;

AVP ; e.g. other corals may have different feeding methods

Different light wavelength penetration with increased depth

[3]

- (ii)** Suggest how the zooxanthellae may benefit in two ways from their association with the corals.

any **two** from:

Physical support to obtain light

Carbon dioxide from coral / respiration, for photosynthesis ;

N from nitrogenous waste of corals / polyps ;

Ref to coral and food caught / suspension feeding / catching preys, provides nutrients / needed for growth of algae ;

Protection from predation ;

Protection from extreme conditions ;

AVP; e.g. low concentrations of ions in seas

[2]

Under conditions of environmental stress the relationship between the reef building corals and the zooxanthellae can break down. Loss of zooxanthellae and the subsequent whitening that occurs is known as coral bleaching. Coral bleaching can lead to death of the coral.

- (b) State **one** reason why permanent loss of zooxanthellae can lead to death of the coral.

any **one** from:

Decreased food source ;

Less organic compounds / named compounds ; @carbon fixation

Loss (of) main source of (chemical) energy ;

Loss of inorganic ions for deposition of coral skeleton ;

AVP; e.g. loss of protective algal layer from harmful effects of sunlight ;

[1]

- (c) One type of environmental stress that can cause coral bleaching is an increase in sea temperature.

- (i) Suggest why areas of sea with reef building corals are particularly susceptible to increased temperature as a result of global climate change.

any **two** from:

Idea that coral reefs grow in shallow / tropical seas ;

Shallow water heats up more rapidly/surface water are warmer than deeper water ;

Temperature increases (due to global warming) may be greater near the equator;

[2]

- (ii) The optimum temperature range for the survival of reef building corals is 25°C to 29°C.

Explain why reef building corals will be affected by an increase in temperature above the optimum.

any **two** from:

Ref to temperature above optimum for enzymes ;

Examples of life processes affected e.g. photosynthesis, respiration, synthesis of biomolecules ;

Ref to other temperature effect ; e.g. damage to membranes ;

[2]

[Total: 10]